

Carlos Durán Cañas

MECHANICAL ENGINEER MSc.

 Carlos Durán Cañas |  Portfolio |  carlosduranc@protonmail.com |  +1 (908) 848-3263

SUMMARY

A dedicated individual driven by genuine passion and hard work to achieve goals. Excelled in fields like thermodynamics, fluid dynamics, and optimal control. Thrives in collaborative environments, adapting seamlessly and contributing with a proactive mindset.

WORK EXPERIENCE

DEVELOPMENT ENGINEER - HEAT PUMP CONTROL

Sep 2023 - Dec 2024

Daikin Europe NV

Ghent, Belgium

- Specialized in developing and testing thermodynamic system controllers for heat pump installations with air, ground source, and water-to-water setup capabilities, delivering space heating, cooling, and domestic hot water.
- Utilized Model-in-the-Loop simulations with MathWorks' Simulink Test to provide crucial input, detailed reports, and design tailored test scripts, ensuring reliability and performance of new system functionalities.
- Collaborated closely with controller designers and unit testers, directly contributing to the implementation of controller logic and quickly assuming significant responsibilities in a dynamic development environment.

ACADEMIC CO-CREATION

Mar 2020 - Jan 2021

Mercedes-Benz Group AG

Sindelfingen, Germany

- Implemented a Semantic Data Model within the Mercedes Operation 360 environment, enabling communication between legacy and evolving data storage systems. This involved extensive research in database technologies, big data processing, and graph technologies, and was successfully pitched to a senior executive on the Board of Management overseeing Production, Quality & Supply Chain Management.
- Awarded the “Problem Solving Heroes” Award for outstanding contributions to the project.

INTERNSHIP PROGRAM

Sep 2019 - Nov 2019

Gildan Honduras Trading S. de R.L.

Rio Nance, Honduras

- Participated in all stages of the steam generation process in a biomass plant within the industrial park, from biomass delivery logistics and quality checks to boiler controls for steam delivery to multiple park plants.
- Executed maintenance procedures and conducted routine furnace rotations for steam generation.
- Established a comprehensive maintenance database for a knitting plant, covering maintenance procedures, safety protocols, schedules, and asset management.

EDUCATION

2021 - 2023 MECHANICAL ENGINEERING MSc.
Thermotechnical Sciences Track

Catholic University of Leuven (KU Leuven)
CGPA: 3.6 - CUM LAUDE

2017 - 2021 MECHANICAL ENGINEERING BSc.
Energy Specialization

Karlsruhe Institute of Technology (KIT)
CGPA: 3.3

PUBLICATIONS

Durán Cañas, Carlos et al. (2023). “Parameter Estimation of Modelica Building Models Using CasADi”. In: *Modelica Conferences*, pp. 301–310. URL: <https://ecp.ep.liu.se/index.php/modelica/article/view/938>.

SKILLS

LANGUAGES

Spanish	English
Dutch	German

SOFTWARE

Python	Matlab	Simulink	Solid Edge	SolidWorks	SimaPro
Git	JIRA	LabVIEW	Ansys Fluid	Modelica	PTC Creo